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|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--|---|--|---|--|---|--|
| 16 | <p>A wave pulse of length 0.25 m is shown at time <math>t = 0</math>. It moves along a string, fixed at point O, at <math>0.50 \text{ m s}^{-1}</math>.</p> <div data-bbox="611 126 1032 367"></div> <p>Which graph best represents the displacement of point P with time for the period <math>t = 1.5 \text{ s}</math> to <math>2.0 \text{ s}</math>?</p>                                                                                                                                                                                                               |   |  |   |  |   |  |   |  |
|    | <table border="1"><tr><td data-bbox="194 519 263 656">A</td><td data-bbox="263 519 486 656"></td><td data-bbox="486 519 555 656">B</td><td data-bbox="555 519 780 656"></td><td data-bbox="780 519 850 656">C</td><td data-bbox="850 519 1074 656"></td><td data-bbox="1074 519 1144 656">D</td><td data-bbox="1144 519 1367 656"></td></tr></table>                                                                                                                                                                                                                     | A |  | B |  | C |  | D |  |
| A  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | B |  | C |  | D |  |   |  |
| L3 | <p><b>Answer: A</b></p> <p>At time <math>t = 1.0 \text{ s}</math>, the pulse will be between P and O, and start to be reflected.</p> <p>At time <math>t = 1.5 \text{ s}</math>, the entire pulse would have reflected, and will be between P and O but the wave will be represented as:</p> <div data-bbox="761 859 876 946"></div> <p>From time <math>t = 1.5 \text{ s}</math> to <math>2.0 \text{ s}</math>, the reflected wave will pass through point P with the above wave representation. So point P will be displaced downwards slightly first, then upwards.</p> |   |  |   |  |   |  |   |  |

**17** A narrow parallel beam of unpolarised light is passed through two optical polarisers. The first